# Inference for measures of agreement with multiple raters

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### Abstract

Most measures of agreement are chance-corrected measures of agreement. They differ in three dimensions: their definition of chance agreement, their choice of disagreement function, and how they handle multiple raters. Chance agreement is usually defined following either Cohen's kappa or Fleiss' kappa. The disagreement function is usually either the nominal, quadratic, or the absolute value function. But how to handle multiple raters is contentious, with the main contenders being Fleiss' kappa, Conger's kappa, and Hubert's kappa, the variant of Fleiss' kappa where agreement is only said to occur if every rater agrees. More generally, multi-rater agreement coefficients can be defined in a $g$-wise way, where the disagreement weighting function uses $g$ of the raters. This paper contains two main contributions. (a) We propose to use Fréchet variances to handle the case of multiple raters. The Fréchet variances are intuitive disagreement measures, and turn out to generalize the nominal, quadratic, and absolute value functions to the case of more than 2 raters. (b) We derive the limit theory of $g$-wise weighted agreement coefficients, with chance agreement either of the Cohen type or Fleiss type, for the case when every item is rated by the same number of raters. Trying out three confidence interval constructions, we end up recommending to calculate confidence intervals using the arcsine transform or the Fisher transform.

## 1 Introduction

Most popular measures of inter-rater agreement involve correction for chance agreement. These can be written on the form

$$\frac{p_a - p_{ca}}{1 - p_{ca}} = 1 - \frac{p_d}{p_{cd}}, \tag{1.1}$$

where $p_a$ ($p_d$) is the percent agreement (disagreement) between the raters and $p_{ca}$ ($p_{cd}$) is the chance agreement (disagreement) between the raters. Such measures are frequently called *chance-corrected measures of agreement*. Well-known examples of coefficients in this class are Cohen's (1960) kappa and its weighted variant (1968), Krippendorff's (1970) alpha, Scott's (1955) pi, and Fleiss' (1971) kappa. Some of these coefficients are either defined only for two raters. The rest are defined in a pairwise manner, in the sense they measure agreement between two raters at a time. But not every proposed measure of agreement is defined on pairs of raters. The most famous is Hubert's kappa (1977), which was recently studied in detail by Martín Andrés and Álvarez Hernández (2020). Other two-rater agreement coefficients include Conger's kappa (Conger, 1980; Light, 1971), which is Cohen's kappa generalized to multiple raters, the $AC_1$ coefficient (Gwet, 2008), and the recent coefficient of van Oest (2019).

There is no consensus about how multi-rater agreement coefficients should be defined. Broadly speaking, there are two options, pairwise coefficients and consensus coefficients. Pairwise coefficients measure agreement between pairs of raters (Conger, 1980), while consensus coefficients measure the simultaneous agreement between all raters. In particular, consensus coefficients support the notion that "agreement occurs if and only if all raters agree on the categorization of an object" (Hubert, 1977). Both pairwise and consensus-based definitions of agreement are variants of $g$-wise measures of agreement (Conger, 1980), where agreement is measured among $g$-tuples of raters. But not-trivial ways to measure agreement is hard to come by when $2 < g < R$. In Section 3 we introduce a framework for handling $g$-wise measures on agreement based on the concept of *Fréchet variances* (Dubey & Müller, 2019). The Fréchet variances, based on the Fréchet mean, generalize the variance, and the measures of agreement based on them generalize the nominal, linearly weighted, and quadratically weighted pairwise measures of agreement in a natural way. They are easily interpretable as, you measure how much do the raters disagree with the (generalized) mean rater, and then adjust for chance. For nominal data in particular, they measure how many raters disagree with the modal rater, with a resulting agreement measure less extreme than Hubert's kappa (Hubert, 1977).

Assuming multivariate normality of the judgments, Lin (1989, Section 3) derived the asymptotic distribution of Cohen's kappa with quadratic weights. Fleiss (1971) introduced a formula for the standard error of Fleiss' kappa, but later showed that it was incorrect. Using the properties of the multinomial distribution and the delta method, Schouten (1982) found the asymptotic variance of the weighted Fleiss' kappa in the case when the number of categories is finite. Almost forty years later, Gwet (2021) found a consistent estimator of the variance for the unweighted Fleiss' kappa. We extend these results to the weighted $g$-wise Fleiss' kappa for any number of categories below. In addition, we mention that bootstrap inference for Fleiss' kappa and Krippendorff's alpha was studied by Zapf, Castell, Morawietz, and Karch (2016).

We begin the paper by providing the definitions of two kinds of chance-corrected agreement coefficients, then we establish connections between a multi-rater weighted Cohen's kappa, the weighted Fleiss' kappa, Conger's kappa, Krippendorff's alpha, and Hubert's kappa (Section 2). We restrict ourselves to the simple context when every rater rates every item. In Section 3 we introduce the Fréchet variances, a useful class of disagreement functions that make sense when $2 < g < R$. Then we spell out the basic limit theory for this class agreement coefficients (Section 4), extending the results of Schouten (1980), Schouten (1982) and O'Connell and Dobson (1984) to vector-valued items and $g$-wise coefficients. We do this using $U$-statistics (Lee, 2019), but there are other ways to arrive at the same results. Then we provide practical recommendations regarding the choice of confidence interval construction, obtained by comparing three confidence interval constructions: basic, arcsine transformed, and Fisher transformed. Using a simulation study (Section 5.1), we find that the arcsine and Fisher intervals outperform the basic interval when $n$ is small.

## 2 Measures of agreement

Discussing measures of inter-rater agreement requires quite a few definitions. Let $d(x_1, \ldots, x_g)$ be a *disagreement function*, a positive and symmetric function of $g$ arguments that equals 0 when all $x_i$s are equal, i.e., $d(x, \ldots, x) = 0$. This function measures the disagreement between the judgments $x_1, \ldots, x_k$, where 0 is understood as complete agreement.

All disagreement functions in widespread use take two arguments. While there are infinitely many disagreement functions, the best known belong to the class of $l_p$ quasi-norms, $p = 0, 1, 2$, potentially raised to the $p$th power. The $l_p$ quasi-norms, $p \in [0, \infty]$ on $\mathbb{R}^g$ are defined as

$$\|x\|_p = (\sum_{i=1}^{g} |x_i|^p)^{1/p}. \tag{2.1}$$

Here $||x||_0 = \sum_{i=1}^{g} 1[x_i \neq 0]$ and $||x||_\infty = (\sum_{i=1}^{g} |x_i|^p)^{1/p}$, as can be verified by taking the limit of $||x||_p$ as $p \to 0$ and $p \to \infty$, respectively. It is well known that $||x||_p$ are proper norms if and only if $p \geq 1$, as the triangle inequality is violated when $1 > p \geq 0$.

Define the disagreement functions $d_p$ as the $l_p$-norm evaluated in $x_1 - x_2$, i.e.,

$$d_p(x_1, x_2) = ||x_1 - x_2||_p. \tag{2.2}$$

In the case of scalar values, $d_0(x_1, x_2) = 1[x_1 \neq x_2]$ is known as the *nominal disagreement function*. For $p = 1$, the $l_p$-norm equals $d_1(x_1, x_2) = |x_1 - x_2|$, which is known as the *absolute value disagreement function* (and sometimes the linear disagreement function). The *quadratic disagreement function* is $d_2^2(x_1, x_2) = (x_1 - x_2)^2$. Vector-valued variants of $d_p$ and $d_p^p$ are far less common, but have been used by e.g. Berry, Johnston, and Mielke (2008)

When the dimension of $d$ isn't equal to 2, we are mostly interested in the case when its arty equals the number of raters $R$. In this case, the disagreement functions often measure the degree of consensus among the raters, with 0 reflecting complete consensus. The most obvious choice is the Hubert disagreement function

$$d(x_1, \ldots, x_K) = 1 - 1[x_1 = \cdots = x_K] \tag{2.3}$$

which equals 0 if and only if every rater agrees on a rating. It is the disagreement function employed in Hubert's kappa (Hubert, 1977). This is not the only reasonable disagreement function for multiple raters, however. We will come back to that in Section 3.

We present our results in terms of disagreement functions instead of the more popular agreement functions (i.e., positive functions bounded by 1 where 1 signifies maximal agreement, sometimes with the additional assumption that $a \geq 0$). We do this mainly for mathematical convenience. Agreement functions and disagreement functions are tightly related, for if $a$ is an agreement function then $d = 1 - a$ is a disagreement function. See the appendix, p. 20 for a more detailed discussion.

Let $R$ denote the number of raters and $n$ denote the numbers of items rated. We make several assumptions. First, we assume that the ratings are conditionally independent given the item. Second, we assume that every item is rated by exactly the same number of raters, which we refer to as the *rectangular design*, which is common in the literature. [1] The rectangular design assumption can be relaxed, but it is needed for the limit results. We sketch how to loosen it up in the appendix (p. 22). Third, we will assume either: (a) Fixed raters, so the same set of raters rate every item, most commonly associated with applications of Cohen's kappa, or (b) Exchangeable ratings given the item, which is the case when the ratings are independent and identically distributed conditional on the item rater, an implicit assumption underlying most applications of Fleiss' kappa, e.g., (Fleiss, 1971). In case (b), the marginal distributions for all raters will be equal, which implies that the population value of the Fleiss' kappa equals the population value of the generalized Cohen's kappa, both defined below. But the sample Fleiss' kappa is the preferred sample estimator, as it is invariant under changes of the raters' identities.

We will also assume that the ratings are made independently of one another in the sense the ratings on the $i$th item in no way affects the ratings on $k$th item. This situation corresponds to a statistical model where $\boldsymbol{X}_1, \boldsymbol{X}_2, ..., \boldsymbol{X}_n$, $R$-dimensional vectors (whose elements may also be vectors) are sampled independently from some distribution $F$. Be aware that we *do not* assume that $\boldsymbol{X}_k$ contains only categorical values, or a finite number of values; the elements of $\boldsymbol{X}_k$ are arbitrary in our framework.

Our intention is to collect the kappas of Cohen, Fleiss, Conger, Hubert, and so on, into a coherent framework of $g$-wise agreement coefficients. To do this we will have to define some quantities. Let $\boldsymbol{x}_i = (x_{i1}, x_{i2}, \ldots, x_{iR})$ be an $R$-dimensional vector of observed ratings and recall that $g$ is the dimension of our disagreement function $d$.

[1] For instance, Fleiss (1971), in his paper introducing Fleiss' kappa, removed several ratings from this data to make sure the total number of ratings was 6 for each item.

(i) **The disagreement $\boldsymbol{x}_1$,** as measured by $d$. The purpose of this quantity is to translate an arbitrary $g$-dimensional disagreement function $d$ into a disagreement function taking an $R$-dimensional vector $\boldsymbol{x}_i$ as input. It is defined as

$$D_d(\boldsymbol{x}_1) = \binom{R}{g}^{-1} \sum_{j_i} d(\boldsymbol{x}_{1j_1}, \dots, \boldsymbol{x}_{1j_g}), \tag{2.4}$$

where the sum runs over all $g$-dimensional subsets of $\{1, \dots, R\}$ with order ignored, i.e., the $g$-combinations of $R$. The expression is simplified when $g = R$, as $D_d(\boldsymbol{x}_1) = d(\boldsymbol{x}_{11}, \dots, \boldsymbol{x}_{1R})$ in this case. To gain some intuition about this quantity, suppose that $g = 2$, that $x_1, x_2$ are scalars, and consider the nominal disagreement function $d_0(x_1, x_2) = 1[x_1 \neq x_2]$. Then $D_d(\boldsymbol{x}_1) = 2R^{-1}(R-1)^{-1} \sum_{j_1 > j_2} 1[x_{1j_1} \neq x_{1j_2}]$ is the percentage of times two raters precisely agree on their rating.

(ii) **The Cohen-type chance disagreement** at $\boldsymbol{x}_1, ..., \boldsymbol{x}_g$, so called to differentiate it from the Fleiss-type chance disagreement. It is similar to the disagreement at $\boldsymbol{x}_1$, but this time the raters do not necessarily rate the same item, as one rater raters the first item (from $\boldsymbol{x}_1$) another rater raters the second item (from $\boldsymbol{x}_2$), and so on. We do not allow a rater to rate the same item more than once in a pass: Hence we need to choose $g$ raters from a set of $R$ raters, and the chance disagreement is

$$C_d(\boldsymbol{x}_1, ..., \boldsymbol{x}_g) = \binom{R}{g}^{-1} \sum_{j_i} d(\boldsymbol{x}_{1j_1}, \dots, \boldsymbol{x}_{gj_g}), \tag{2.5}$$

where the sum runs over all $g$-dimensional subsets of $\{1, \dots, R\}$, i.e., the $g$-combinations of $R$. Observe that $D_d(\boldsymbol{x}) = C_d(\boldsymbol{x}, , \dots, \boldsymbol{x})$. Since $d$ is assumed to be symmetric, the expression is massively simplified to $d(\boldsymbol{x}_{1j_1}, \dots, \boldsymbol{x}_{Rj_R})$ when $g = R$. When $g = 2$, $C_d(\boldsymbol{x}_1, \boldsymbol{x}_2) = R^{-1}(R-1)^{-1} \sum_{j_1 \neq j_2} d(\boldsymbol{x}_{1j_1}, \boldsymbol{x}_{2j_2})$.

(iii) **The Fleiss-type chance disagreement** at $\boldsymbol{x}_1, ..., \boldsymbol{x}_g$ is similar, but allows the same rater (or rater index) to rate an item multiple times. Its definition is

$$F_d(\boldsymbol{x}_1, ..., \boldsymbol{x}_g) = R^{-g} \sum_{j_i} d(\boldsymbol{x}_{1j_1}, \dots, \boldsymbol{x}_{gj_g}),$$

where the sum runs over the product set $R^g$. The expression for $F_w(\boldsymbol{x}_1, ..., \boldsymbol{x}_g)$ is not dramatically simplified when $g = R$. When $g = 2$, $F_d(\boldsymbol{x}_1, \boldsymbol{x}_2) = R^{-2} \sum_{j_1, j_2} d(\boldsymbol{x}_{1j_1}, \boldsymbol{x}_{2j_2})$.

We will call the expected values of these quantities the *mean disagreement*, the *mean Cohen-type chance disagreement*, and the *mean Fleiss-type chance disagreement*. Slightly abusing notation, we denote them as

$$D_d = E[D_d(\boldsymbol{X}_1)], \quad C_d = E[C_d(\boldsymbol{X}_1, \dots, \boldsymbol{X}_g)], \quad F_d = E[F_d(\boldsymbol{X}_1, \dots, \boldsymbol{X}_g)], \tag{2.6}$$

where $\boldsymbol{X}_1, \dots, \boldsymbol{X}_g$ are independently sampled from the same distribution $F$. Discussions about the difference between $E[C_d(\boldsymbol{X}_1, \dots, \boldsymbol{X}_g)]$ and $E[F_d(\boldsymbol{X}_1, \dots, \boldsymbol{X}_g)]$, and why to prefer one over the other, are abundant in the literature, often in the context of the so-called paradox of kappa (Cicchetti & Feinstein, 1990).

**Definition 1.** Let $X \sim F$ be a vector of $R$ ratings and $d$ be an agreement function with dimension $g$. Define the population values of the generalized Cohen's kappa ($\kappa_d$) and Fleiss' kappa ($\pi_d$) as

$$\kappa_d = 1 - \frac{D_d}{C_d}, \quad \pi_d = 1 - \frac{D_d}{F_d}. \tag{2.7}$$

Table 1: Weighted agreement coefficients

| **Coefficient** | $R = 2$ | $R > 2$ | |
|---|---|---|---|
| | | $g = 2$ | $g = R$ |
| Cohen-type ($\kappa_d$) | Cohen's kappa<br>Lin's CC* | Conger's kappa† | Schuster–Smith |
| Fleiss-type ($\pi_d$) | Scott's $\pi$† | Fleiss' kappa†<br>Krippendorff's alpha | Hubert's kappa† |

*Lin's concordance coefficient is defined for quadratic weights only.
†Originally defined for nominal weights only.

We have decided to denote Fleiss' kappa as $\pi_d$ as it's strictly speaking a generalization of Scott's pi (Scott, 1955), not Cohen's kappa. When $g = 2$, the kappas are commonly known as pairwise agreement coefficients. When $2 \leq g \leq R$, they are called $g$-wise agreement coefficients, and when $g = R$, they are sometimes called consensus coefficients. The case when $2 < g < R$ has received little attention in the literature, with some exceptions (Warrens, 2012).

The definition of the weighted Fleiss' kappa is a straight-forward generalization of Fleiss' kappa (1971) to hold for any disagreement function of any dimension, not just $g = 2$. When $g = R$ and $d$ is the nominal disagreement, it equals Hubert's kappa. Likewise, the definition of the weighted Conger's kappa is an extension of Conger's kappa to the weighted case with $g \neq 2$ allowed. When $g = R$, it equals the Schuster–Smith coefficient (Schuster & Smith, 2005, eq. 1)[2]. Moreover, Berry and Mielke (1988) discussed $\kappa_d$ for Euclidean weights between vector-valued ratings, while Janson and Olsson (2001) extended it to squared Euclidean weights and nominal weights. The relationship between most of the agreement coefficients mentioned until now is summarized in Table 1.

## 3 Fréchet variances for $g$-wise agreement coefficients

The most popular measures of agreement are defined only in a pairwise manner with $g = 2$. It is easy to find reasonable disagreement measures in this case, as one can draw on the literature on norms and distances. The $l_p$ distances are the obvious choices, but there are many other options, such as the Huber loss (Huber, 1964) or LINEX loss (Varian, 1975).

In the setting of Hubert's kappa and the Schuster–Smith coefficient we have $g = R$, and it's suddenly not that easy to find reasonable disagreement functions anymore. The disagreement weight used in Hubert's kappa, $d(x_1, \ldots, x_R) = 1 - 1[x_1 = \cdots = x_R]$, will penalize any number of discordant ratings equally, yielding the often undesirable outcome that most sets of ratings will be in complete disagreement. But there are less sensitive way to count nominal disagreements. Consider the case of 10 raters using a nominal scale from $1 - 3$, with 7 giving rating 1, 2 giving rating 2, and 1 giving rating three. Then Hubert's disagreement rating is 1, as there are raters who disagree at all, and the pairwise (nominal) disagreement is $46/100$. But it sounds reasonable to pick the modal rating (in this case 1), then report the number of ratings that disagree, in this case 3, divided by the number of raters. This "modal" disagreement equals $3/10$.

Sometimes we wish to report the aggregate judgment from a set of raters. For instance, the median judgment of an item could be used as input to a regression model. In this case it's natural to measure how much more

[2] The Schuster–Smith coefficient, a generalized Cohen's kappa, encompasses the case of $2 < g < R$ provided their weight function $v(s)$ is appropriately defined, see the discussion on dispersion weights in (Schuster & Smith, 2005).

we get to know about the median of an item by asking our raters, compared to the situation when ask no one. Consider the case above again, but using the median instead of the mode. The aggregate judgment would be the median judgment, which equals 1. It is well-known that the median of a vector $x$ equals $\text{argmin}_\mu \sum |x_i - \mu|$, so this appear to be a reasonable measure of mean disagreement when we use the median as aggregation method.

The "modal" disagreement and the $\text{argmin}_\mu \sum |x_i - \mu|$ disagreement are instances of an intuitive generalization of the variance called the (generalized) Fréchet variance (Dubey & Müller, 2019). The Fréchet mean (Fréchet, 1948) is a generalization of centroids to arbitrary distance functions $l$[3]; likewise, the Fréchet variance is a generalization of dispersion to any such distance function. Let $l$ be a distance function satisfying $l(x, y) \geq 0$ and $l(x, x) = 0$, and let $A = \{x_1, x_2, \ldots, x_n\}$ be a set of points. The sample *Fréchet mean* of $A$ is defined as the (not necessarily unique) point $\mu_l$ that minimizes the sum of distances to all points in $A$, i.e.

$$\mu_l = \text{argmin}_\mu \sum_{k=1}^{n} l(\mu, x_k). \tag{3.1}$$

Similarly, the sample *Fréchet variance* on $A$ with loss function $l$ is

$$V(l) = \min_\mu \sum_{i=1}^{n} \frac{1}{n} l(\mu, x_i) = \sum_{k=1}^{n} \frac{1}{n} l(\mu_l, x_i). \tag{3.2}$$

The Fréchet mean and Fréchet variance are best understood through a decision-theoretic lens: The Fréchet mean of $A$ represents your best judgment of an item's true classification or value according to the loss $l$; the Fréchet variance $V(l)$ is the decision-theoretic risk associated with the choice.

Define the $g$-dimensional disagreement based on $l$ as

$$d(\boldsymbol{x}_1, \ldots, \boldsymbol{x}_g) = V(l)[\boldsymbol{x}_1, \ldots, \boldsymbol{x}_g]. \tag{3.3}$$

Now we may use the definitions of Cohen's kappa and Fleiss' kappa to calculate $\kappa_{d'}$ and $\pi_{d'}$.The disagreement measure is easy to understand and visualize, and corresponds to our intuitive understanding of disagreement quite well.

We will consider the following distances:

(i) $d_0(x, y) = 1[x \neq y]$**.** Generalizes the nominal distance. If the data is categorical, the generalized Fréchet mean $\mu_d$ equals the mode, and the generalized Fréchet variance equals the percentage of observations different from the mode. If we are dealing with vector-valued data with $I$ elements each, it might be preferable to use $\sum_{i=1}^{I} 1[x_i \neq y_i]$ instead, as it counts each dimension of the nominal data separately.

(ii) $d_1(x, y) = ||x - y||_1$. The generalized Fréchet mean equals the coordinate-wise sample median. The generalized Fréchet variance equals the coordinate sum of sample mean absolute deviation from the median, i.e., $\sum_{i=1}^{i} \frac{1}{R} \sum_{k=1}^{R} |x_{ri} - \mu_i|$, where $\mu_i$ are the coordinate wise medians.

(iii) $d_2^2(x, y) = ||x - y||_2^2$. The generalized Fréchet mean equals the coordinate wise sample mean $\mu_d = \frac{1}{n} \sum_{i=1}^{n} x_i$, and the Fréchet variance equals the biased sample variance of $\{x_1, x_2, \ldots, x_n\}$, i.e., $V_d = \frac{1}{n} \sum_{i=1}^{n} (x_i - \mu_d)^2$.

(iv) $d_2(x, y) = ||x - y||_2$. The generalized Fréchet mean has no simple formula, but is known as the *geometric median*. If $x_1$ and $x_2$ are scalars, $d_2 = d_1$, which implies that the Fréchet mean equals

[3] The Fréchet mean and variances is usually defined slightly differently, using $l^2(x, x_k)$ instead of $l(x, x_k)$, with $l$ being a metric. Our definition of the Fréchet mean is sometimes called the generalized Fréchet mean or the $\alpha$-Fréchet mean.

the median, hence the name. There is an extensive literature about the geometric median, see e.g. Drezner, Klamroth, Schöbel, and Wesolowsky (2002) for an overview and M. B. Cohen, Lee, Miller, Pachocki, and Sidford (2016) for how to compute it. Be aware that the geometric median is computationally more intensive than the generalized Fréchet mean based on $||x-y||_2^2$.

For any $p \in [0, \infty]$ and pair of vectors $x_1, x_2$, it holds that

$$V(d_p)[x_1, x_2] = \frac{1}{2} d_p(x_1, x_2), \quad V(d_p^p)[x_1, x_2] = \frac{1}{2^p} d_p^p(x_1, x_2). \tag{3.4}$$

It follows that $\kappa_{d_p} = \kappa_{V(d_p)}$ and $\kappa_{d_p^p} = \kappa_{V(d_p^p)}$ when we are dealing with pairwise agreement. Thus the Fréchet variances generalize the pairwise agreement for these distances (proved in the appendix, page 21).

**Example 2.** Suppose you have $R = 5$ raters rater $4$ items, yielding ratings $(1,1,2,1,1)$, $(1,2,3,2,2)$, $(2,1,1,1,1)$, $(2,3,4,4,5)$. The Fréchet means using the loss $l(x,y) = d_1(x,y) = |x-y|$ equals the coordinate-wise sample medians $1,2,1,4$. The Fréchet variances are $V(d_1) = (0.2, 0.4, 0.2, 0.8)$. To calculate the sample Cohen's kappa with $d = V(d_1)$ we first find the mean disagreement (4.1) $\overline{V(d_1)} = 0.4$, then the mean Cohen disagreement (4.2), which is $\approx 0.73$. Thus Cohen's kappa is $1 - 0.4/0.73 = 0.45$.

We believe the most useful distance measures will typically be $d_0$ and $d_1$, where $d_0$ would be used when there is no known ordinal structure in the data, and $d_1$ would be used if there is a known ordinal structure. The quadratic distance $d_2^2$ could be used as well, but is harder to justify, as it's usually not obvious why we would be interested in the squared distance between two observations rather than just the distance itself. The distances $d_p, p \in (1, \infty]$, with $d_2$ included, are even harder to recommend, as they do not work in a coordinate-wise manner. In any case, it seems most reasonable to go with the $R$-wise variants of these distance measures, as they make use of all the available information, but $g$-wise agreement coefficients $(g < R)$ do not.

The main reason to use Fréchet variances is its simple and practical interpretation. Consider the case of nominal ratings with $R$ raters. You might have to supply a final aggregated rating from all of your ratings, and then it is very natural to use the mode. The Fréchet variance using $d_0$ tells you how much better your aggregate prediction based on the mode will be, compared to using just the marginal distribution of the ratings.

**Example 3** (Fleiss (1971) )**.** In the paper introducing what is now called Fleiss' kappa, Fleiss (1971) discussed an example involving 5 different kinds of diagnoses, $n = 30$ patients, and $R = 6$ psychiatrists. The data is originally from Sandifer, Hordern, Timbury, and Green (1968), but Fleiss removed some ratings to make the design rectangular. The raters are not the same for all raters, but it appears plausible to assume that ratings are exchangeable given the item. The diagnoses are essentially categorical in nature, hence we will only consider $V(d_0)$ and Hubert's disagreement function. When $g = R = 6$, the difference between Hubert's kappa ($\pi = 0.166$) and Fleiss' kappa using the $V(d_0)$ distance ($\pi = 0.486$) is substantial, which suggests that a sizeable number of rating vectors contain at least one rating that disagrees with the others. Table 2 summarizes the relevant aspects of the data. The maximal agreement row could potentially go from $1$ to $6$, but on this data set, the smallest number raters agreeing on a classification is $3$. According to the Hubert distance the raters disagree quite a lot, as only $5$ items have a distance of $0$ and the rest a distance of $1$. On the other hand, $V(d_0)$ equals $6$ minus the maximal agreement (i.e., the number of elements distinct from the mode), and results in much smaller chance-adjusted agreement.

Table 2: Maximal agreement for the data of Fleiss (1971)

| | | | | |
|---|---|---|---|---|
| Maximal agreement* | 3 | 4 | 5 | 6 |
| Count | 8 | 10 | 7 | 5 |
| Distance ($V(d_0)$) | 3 | 2 | 1 | 0 |
| Distance (Hubert) | 1 | 1 | 1 | 0 |

*The largest number of raters that agree on the classification of an item. Both $V(d_0)$ and Hubert's distance depend only on this when $g = R$.

## 4 Estimation and inference

### 4.1 Sample estimates

Let $\boldsymbol{X}_1, ..., \boldsymbol{X}_n \sim F$ be $n$ iid vectors of ratings by $R$ raters. Then there is a single natural sample estimator of $D_d$, namely

$$\hat{D}_d = n^{-1} \sum_i D_d(\boldsymbol{x}_i). \tag{4.1}$$

There are, however, two natural estimators of the Cohen-type chance disagreement, one them a $V$-statistic (Lee, 2019, Chapter 4.2), the other a $U$-statistic (Lee, 2019, Chapter 1),

$$\hat{C}_d \quad = \quad n^{-g} \sum_{i_k} C_d(\boldsymbol{x}_{i_1}, ..., \boldsymbol{x}_{i_g}) \quad \text{and} \quad \hat{C}_d^u = \binom{n}{g}^{-1} \sum_{i_k} C_d(\boldsymbol{x}_{i_1}, ..., \boldsymbol{x}_{i_g}), \tag{4.2}$$

where the first estimator counts every combination with repetitions of $i_1, i_2, \dots, i_g$ and the second estimator counts the unordered combinations $i_1 < i_2 < \dots < i_g$. Using the basic results of $U$-statistics (Lee, 2019, Chapter 1), we see that $C_d^u$ is the unique minimum variance unbiased estimator of $C_d$. Moreover, from a well-known correspondence between $U$-statistics and $V$-statistics, the asymptotic distributions of $\hat{C}_d$ coincides with the asymptotic distribution of $\hat{C}_d^u$ (Lee, 2019, Chapter 4, Theorem 1), so the choice of between $\hat{C}_d$ and $\hat{C}_d^u$ barely matters when $n$ is sufficiently large.

Likewise, there are two natural estimators of the Fleiss-type weighted chance agreement,

$$\hat{F}_d \quad = \quad n^{-g} \sum_{i_k} F_d(\boldsymbol{x}_{i_1}, ..., \boldsymbol{x}_{i_g}) \quad \text{and} \quad \hat{F}_d^u = \binom{n}{g}^{-1} \sum_{i_k} F_d(\boldsymbol{x}_{i_1}, ..., \boldsymbol{x}_{i_g}), \tag{4.3}$$

where the index sets are as described above.

Now we can define two sample variants of Cohen's kappa (Fleiss' kappa), depending on which one of $\hat{C}_d$ ($\hat{F}_d$) and $\hat{C}_d^u$ ($\hat{F}_d^u$) we choose to use. These are $\hat{\kappa}_d = 1 - \hat{D}_d/\hat{C}_d$ and $\hat{\kappa}_d^u = 1 - \hat{D}_d/\hat{C}_d^u$ for Cohen's kappa and $\hat{\pi}_d = 1 - \hat{D}_d/\hat{F}_d$ and $\hat{\pi}_d^u = 1 - \hat{D}_d/\hat{F}_d^u$ for Fleiss' kappa. The definition of the sample Cohen's kappa (J. Cohen, 1968) agrees with $\hat{\kappa}_w$, not $\hat{\kappa}_w^u$. Likewise, the sample Fleiss' kappa – with nominal disagreement – has definition agreeing with $\hat{\pi}_w$ (Fleiss, 1971). Moreover, $\hat{\pi}_w$ and $\hat{\kappa}_w$ are faster to compute when the data isn't continuous. Since the estimators are asymptotically equivalent in any case, we will stick to the $V$-statistics $\hat{\kappa}_w$ and $\hat{\pi}_w$for estimation, but use the $U$-statistic form when deriving limit distributions.

### 4.2 Limit theory using $U$-statistics

Let $\boldsymbol{X}_1, \ldots, \boldsymbol{X}_n$ be independently and identically distributed and $\psi(x_1, \ldots, x_k)$ be a symmetric function. A $U$-statistic of order $k$ with kernel $\psi$ is

$$U_n = \binom{n}{k}^{-1} \sum_{i_1,\ldots,i_k} \psi(\boldsymbol{X}_{i_1}, \ldots, \boldsymbol{X}_{i_k}), \tag{4.4}$$

where the sum extends of over all $k$-dimensional tuples satisfying $1 \leq i_1 < i_2 < \ldots \leq n$.

The theory of $U$-statistics was established by Hoeffding (1992); for an introduction, see e.g. Chapter 6.1 of Lehmann (2004), Chapter 5 of Serfling (1980), or the textbook of Lee (2019). These references handle $U$-statistics where the $X_i$s are real-valued, but their results, including the simple results below, hold for vector-valued $X_i$s as well (Korolyuk & Borovskich, 2013).

The Fleiss-type (Cohen-type) weighted chance agreement is a $U$-statistic with kernel $F_d$ ($C_d$), of order $g$. The disagreement is a $U$-statistic with kernel $D_d$, which has order 1. To find the asymptotic variance of the kappas, we will use formulas for the asymptotic covariance of $U$-statistics. Let $U_{1n}$ and $U_{2n}$ be two $U$-statistics of $n$ observations with symmetric kernel functions $\psi_1, \psi_2$ of dimension $k_1$ and $k_2$. Define

$$\begin{aligned}
\sigma_1^2 &= \operatorname{Var}(E[\psi_1(\boldsymbol{X}_1, \ldots, \boldsymbol{X}_{k_1}) \mid \boldsymbol{X}_1)]), \\
\sigma_{11}^2 &= \operatorname{Cov}(E[\psi_1(\boldsymbol{X}_1, \ldots, \boldsymbol{X}_{k_1}) \mid \boldsymbol{X}_1)], E[\psi_2(\boldsymbol{X}_1, \ldots, \boldsymbol{X}_{k_2}) \mid \boldsymbol{X}_1)]).
\end{aligned}$$

Then it holds that $n \operatorname{Cov}(U_{1n}, U_{2n}) \to k_1 k_2 \sigma_{11}^2$ and $n \operatorname{Var}(U_{1n}) \to k_1^2 \sigma_1^2$ (Lee, 2019, Theorem 2, p. 76)). It is also possible to calculate the exact covariances, which could potentially make the asymptotic variances for the kappas perform better. See the appendix p. 22 for the formula for the exact covariance (Lee, 2019, Theorem 2, p. 17)).

**Lemma 4.** *Assume every rater rates every item exactly once. Define the parameter vectors* $\boldsymbol{p} = (D_d, C_d, F_d)$ *and* $\hat{\boldsymbol{p}} = (\hat{D}_d, \hat{C}_d, \hat{F}_d)$. *Then* $\sqrt{n}(\hat{\boldsymbol{p}} - \boldsymbol{p}) \xrightarrow{d} N(0, \Sigma)$, *where* $\Sigma$ *is the covariance matrix with elements*

$$\begin{aligned}
\sigma_{11} &= \sigma_D^2 = \operatorname{Var} D_d(\boldsymbol{X}_1), & \sigma_{12} &= \sigma_{CD} = g \operatorname{Cov}(D_d(\boldsymbol{X}_1), \mu_C(\boldsymbol{X}_1)), \\
\sigma_{22} &= \sigma_C^2 = g^2 \operatorname{Var} \mu_C(\boldsymbol{X}_1), & \sigma_{13} &= \sigma_{FD} = g \operatorname{Cov}(D_d(\boldsymbol{X}_1), \mu_C(\boldsymbol{X}_1)), \\
\sigma_{33} &= \sigma_F^2 = g^2 \operatorname{Var} \mu_F(\boldsymbol{X}_1), & \sigma_{23} &= \sigma_{CF} = g \operatorname{Cov}(D_d(\boldsymbol{X}_1), \mu_F(\boldsymbol{X}_1)).
\end{aligned}$$

*Here the variable* $\mu_C(\boldsymbol{X}_1)$, *and* $\mu_F(\boldsymbol{X}_1)$ *are defined as*

$$\mu_C(\boldsymbol{X}_1) = E[C_d(\boldsymbol{X}_1, \ldots, \boldsymbol{X}_g) \mid \boldsymbol{X}_1] \quad \mu_F(\boldsymbol{X}_1) = E[F_d(\boldsymbol{X}_1, \ldots, \boldsymbol{X}_g) \mid \boldsymbol{X}_1].$$

The form of the covariance matrix follows from the remarks preceding the lemma. Asymptotic normality follows from a general theorem about asymptotic normality of $U$-statistics, see e.g. Theorem 2 of Lee (2019, p. 76).

We want to use Lemma 4 to find the limit distribution of the generalized Cohen's kappa and Fleiss' kappa. To this end, recall the multivariate delta method (see, e.g., Lehmann, 2004, Theorem 5.2.3). Let $f : \mathbb{R}^k \to \mathbb{R}$ be continuously differentiable at $\theta$ and suppose that $\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, \Sigma)$. Then

$$\sqrt{n}[f(\hat{\theta}) - f(\theta)] \xrightarrow{d} N(0, \nabla f(\theta)^T \Sigma \nabla f(\theta)), \tag{4.5}$$

where $\nabla f(\theta)$ denotes the gradient of $f$ at $\theta$.

In the case of Cohen's kappa and Fleiss' kappa, we find that

$$\nabla \kappa_d = \frac{1}{C_d}\left(-1, \frac{D_d}{C_d}\right), \quad \nabla \pi_d = \frac{1}{F_d}\left(-1, \frac{D_d}{F_d}\right). \tag{4.6}$$

Using some algebra, the expressions for the asymptotic variances follow from Lemma 4 and the gradients above.

**Proposition 5.** *Assume every rater rates every item exactly once. Then Cohen's kappa $\hat{\kappa}_d$ and $k$-wise Fleiss' kappa $\hat{\pi}_d$ are asymptotically normal, and their asymptotic variances are*

$$\begin{aligned}\sigma_\kappa^2 &= \sigma_D^2\frac{1}{C_d^2} - 2\sigma_{CD}\frac{D_d}{C_d^3} + \sigma_C^2\frac{D_d^2}{C_d^4}, \\ \sigma_\pi^2 &= \sigma_D^2\frac{1}{F_d^2} - 2\sigma_{FD}\frac{D_d}{F_d^3} + \sigma_F^2\frac{D_d^2}{F_d^4}.\end{aligned} \tag{4.7}$$

*These results hold for $\hat{\kappa}_d^u$ and $\hat{\pi}_d^u$ too. Since the sample Krippendorff's alpha equals $\hat{\alpha}_d = \hat{\pi}_d + \frac{1}{2Rn}(1-\hat{\pi}_d)$, it is also asymptotically normal with asymptotic variance $\sigma_\pi^2$.*

With $g = 2$ and a finite number of categories, Schouten (1980) derived $\sigma_\pi^2$, while Schouten (1982) and O'Connell and Dobson (1984) derived $\sigma_\kappa^2$. The result for Krippendorff's alpha is, to our knowledge, new.

**A brief aside on Krippendorff's alpha** Krippendorff's alpha (Krippendorff, 1970) is another popular agreement coefficient, especially popular in content analysis (Krippendorff, 2018). It has no population definition, but its sample definition equals $\hat{\alpha}_d = \hat{\pi}_d + \frac{1}{N}(1-\hat{\pi}_d)$, (the total sample size $N$ equals $2Jn$ in the case of a rectangular design) see Proposition 10 in the appendix for a justification. For this reason, all of the results about the limit of $\hat{\pi}_w^u$ applies to Krip\pendorff's alpha as well, as it is an asymptotically equivalent estimator of $\pi_d$. Note, however, that Krippendorff (2018) emphasizes the use of non-rectangular designs, and the limit results in the preceding section do not hold for such study designs.

### 4.3 Estimating the variances

The unknown disagreement measures $\hat{D}_d$, $\hat{C}_d$ and $\hat{F}_d$ can be estimated using their sample counterparts. The variances and covariances can be estimated using the empirical (co)variances of the estimated $\hat{\mu}$s, defined as

$$\begin{aligned}\hat{\mu}_d(\boldsymbol{x}_i) &= D_d(\boldsymbol{x}_i), \\ \hat{\mu}_{dc}(\boldsymbol{x}_i) &= n^{-(g-1)}\sum_{i_j}^{n} C_d(\boldsymbol{x}_i, \boldsymbol{x}_{i_1}, \ldots, \boldsymbol{x}_{i_{k-1}}), \\ \hat{\mu}_{df}(\boldsymbol{x}_i) &= n^{-(g-1)}\sum_{i_j}^{n} F_d(\boldsymbol{x}_i, \boldsymbol{x}_{i_1}, \ldots, \boldsymbol{x}_{i_{k-1}}).\end{aligned} \tag{4.8}$$

From the definitions of $\hat{D}_d, \hat{C}_d$, and $\hat{F}_d$, (equation 4, p. 8), we quickly deduce that $\overline{\hat{\mu}_d} = \hat{D}_d$, $\overline{\hat{\mu}_{dc}} = \hat{C}_d$ and , $\overline{\hat{\mu}_{fc}} = \hat{F}_d$. Using this fact, we can define the estimators

$$\hat{\sigma}_C^2 = \frac{4}{n-1}\sum_{i=1}^{n}(\hat{\mu}_{dc}(\boldsymbol{x}_i) - \hat{C}_d)^2, \quad \hat{\sigma}_{CD}^2 = \frac{2}{n-1}\sum_{i=1}^{n}(\hat{\mu}_{dc}(\boldsymbol{x}_i) - \hat{C}_d)(\hat{\mu}_d(\boldsymbol{x}_i) - \hat{D}_d),$$

and $\hat{\sigma}_D^2 = \frac{1}{n-1}\sum_{i=1}^{n}(\hat{\mu}_d(\boldsymbol{x}_1) - \hat{D}_d)^2$. Moreover, we can estimate $\hat{\sigma}_F^2$ and $\hat{\sigma}_{FD}^2$ in the same way, substituting $\hat{\mu}_{df}$ for $\hat{\mu}_{dc}$. Using the formulas for the theoretical variances 4.7 we find the estimators

$$\hat{\sigma}_\kappa^2 = \hat{\sigma}_D^2\frac{1}{\hat{C}_d^2} - 2\hat{\sigma}_{CD}\frac{\hat{D}_d}{\hat{C}_d^3} + \hat{\sigma}_C^2\frac{\hat{D}_d^2}{\hat{C}_d^4}, \tag{4.9}$$

$$\hat{\sigma}_\pi^2 = \hat{\sigma}_D^2\frac{1}{\hat{F}_d^2} - 2\hat{\sigma}_{FD}\frac{\hat{D}_d}{\hat{F}_d^3} + \hat{\sigma}_F^2\frac{\hat{D}_d^2}{\hat{F}_d^4}. \tag{4.10}$$

The variance estimator $\hat{\sigma}_\pi^2$ coincides with that of Gwet (2021, equation 4), in the case of nominal weights.

### 4.4 Improving approximate normality with the arcsine and Fisher transforms

It is well-known that the *Fisher transform* (Fisher, 1915) improves inference for the correlation coefficient. If $r$ is the sample correlation, $\operatorname{artanh}(r) = \frac{1}{2}\log[(1+r)/(1-r)]$ has approximately the same variance for most $r$, and its distribution is closer to normal than that of the untransformed $r$, especially when $\rho$ is close to $\pm 1$. This transform makes sense outside the world of correlations; for instance, Lin (1989) used the Fisher transform to improve normality of the quadratically weighted Cohen's kappa.

The arcsine is another reasonable transformation of $\hat{\kappa}_d$ and $\hat{\pi}_d$. The arcsine is the inverse of the sine function from trigonometry, and is defined as $\arcsin x = \int 1/\sqrt{1-x^2}dx$. In ecology (Warton & Hui, 2011), the arcsine transformation denotes $\arcsin\sqrt{p}$, where $p$ is a probability. We do not take square root, however, as $\hat{\kappa}_d$ and $\hat{\pi}_d$ can be negative.

The arcsine is defined for $\hat{\kappa}_d \in [-1,1]$, and the Fisher transform for $\hat{\kappa}_d \in (-1,1)$, and $\hat{\kappa}_d < -1$ is theoretically possible when $d$ isn't quadratic. But, seeing as $\hat{\kappa}_d < -1$ rarely happens, and that inference is of limited interest in such situations of extreme disagreement, we choose to ignore this problem.

**Example 6.** This example illustrates that the arcsine and Fisher transform may make the sampling distribution closer to the normal distribution. Let the number of raters be $R = 3$, the disagreement function take $g = 2$ arguments, and the number of items be $n = 20$. The are five categories, and the true classification of an item is $c = 1,2,3,4,5$ with probability $1/5$ each. Every rater knows the true classification of an item with probability $0.9$. If they to do not know the correct classification, they will guess a classification from $c = 1,2,3,4,5$ uniformly at random. We use quadratic weights. One can show that the population value of the quadratically weighted Cohen's kappa is $0.816$ under these circumstances, following the arguments of Perreault and Leigh (1989). We simulate the value of $\hat{\kappa}_d$ a total of $N = 50,000$ times and transform them using the identity transform, the arcsine transform, and the Fisher transform. The results are in 4.1. The arcsine transform makes appears to make the sampling distribution of $\hat{\kappa}_w$ closest to the normal distribution, with the Fisher transform improving normality quite a bit as well.

Calculating the limiting variance of $\arcsin\hat{\kappa}_d$ and $\arcsin\hat{\pi}_d$ requires an additional application of the delta method. We use that $\frac{d}{dx}\arcsin(x) = 1/\sqrt{1-x^2}$ and $\frac{d}{dx}\operatorname{artanh}(x) = 1/(1-x^2)$. The one-dimensional delta method states that $\sqrt{n}(\hat{\theta}-\theta) \xrightarrow{d} N(0,\sigma^2)$ implies that $\sqrt{n}(f(\hat{\theta})-f(\theta)) \xrightarrow{d} N(0, f'(\theta)^2\sigma^2)$. Hence

$$\sqrt{n}(\arcsin\hat{\kappa}_d - \arcsin\kappa_d) \quad\rightarrow\quad N(0,(1-\kappa_d^2)^{-1}\sigma_\kappa^2), \tag{4.11}$$

$$\sqrt{n}(\operatorname{artanh}\hat{\kappa}_d - \operatorname{artanh}\kappa_d) \quad\rightarrow\quad N(0,(1-\kappa_d^2)^{-2}\sigma_\kappa^2). \tag{4.12}$$

The expressions for $\hat{\pi}_d$ can be found by swapping $\kappa_d$ for $\pi_d$ and $\sigma_\kappa^2$ for $\sigma_\pi^2$.

### 4.5 Examples

**Example 7** (Fleiss (1971) example (cont.))**.**

Fleiss (1971), in the paper introducing what is now called Fleiss' kappa, discussed an example involving 5 different kinds of diagnoses, $n = 30$ patients, and $R = 6$ psychiatrists. The data is originally from Sandifer et al. (1968), but Fleiss removed some ratings to make the design rectangular. The raters are not the same for all raters, but it appears plausible to assume that ratings are exchangeable given the item. The diagnoses are essentially categorical in nature, hence we will only consider $V(d_0)$ and Hubert's disagreement function. The results are shown in Table 3. We see that the agreement coefficients agree when $g = 2$, as both $V(d_0)$ and Hubert's disagreement function equals the nominal agreement in this case. But the coefficients differ substantially as $g$ increases. This is to be expected, as Hubert's disagreement function measures consensus while $V(d_0)$ measures the number of observations different from the mode.

When $g = R = 6$, the difference between Hubert's kappa $(0.166)$ and $\pi_{V(d_0)} = 0.486$ is substantial, which suggests that a sizeable number of rating vectors contain at least one rating that disagrees with the others.

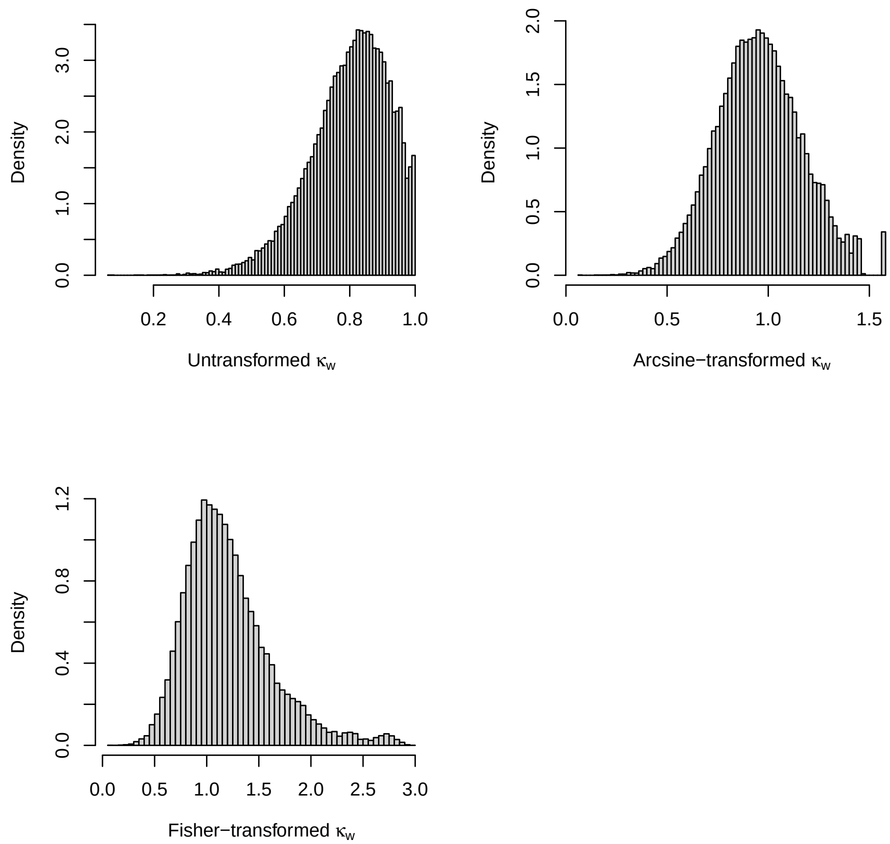


Figure 4.1: Simulated sampling distribution of $\hat{\kappa}_w$ for quadratic weights using three transformations, $n = 20$, $R = 3$. The simulation setup is described in Example 6. The arcsine transform makes the sampling distribution closest to the normal distribution.

Table 3: Confidence intervals for the data of Fleiss (1971) using the arcsine method

| ***d*** | **Fleiss' kappa** | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | $g = 2$ | | | $g = 3$ | | | $g = 6^*$ | | |
| | $\text{CI}_l$ | $\text{CI}_u$ | $\hat{\pi}$ | $\text{CI}_l$ | $\text{CI}_u$ | $\hat{\pi}$ | $\text{CI}_l$ | $\text{CI}_u$ | $\hat{\pi}$ |
| $V(d_0)$ | 0.444 | 0.671 | 0.562 | 0.388 | 0.597 | 0.496 | 0.366 | 0.597 | 0.486 |
| Hubert$^\dagger$ | 0.444 | 0.671 | 0.562 | 0.202 | 0.458 | 0.333 | 0.021 | 0.308 | 0.166 |

$^*$This is Hubert's kappa when the Hubert disagreement is used.
$^\dagger$ Hubert disagreement equals the nominal disagreement $V(d_0)$ when $g = 2$.

Table 4: Confidence intervals for the data of Fleiss (1971) using the arcsine method

| | | | | |
|---|---|---|---|---|
| Maximal agreement* | 3 | 4 | 5 | 6 |
| Count | 8 | 10 | 7 | 5 |
| Distance ($V(d_0)$) | 3 | 2 | 1 | 0 |
| Distance (Hubert) | 1 | 1 | 1 | 0 |

*The largest number of raters that agree on the classification of an item. Both $V(d_0)$ and Hubert's distance depend only on this when $g = R$.

Table 5: Confidence intervals for Zapf et al. (2016) using the arcsine method

| ***d*** | **Cohen's kappa** | | | | | | **Fleiss' kappa** | | |
|---|---|---|---|---|---|---|---|---|---|
| | $g = 2$ | | | $g = 4^{\dagger}$ | | | $g = 4$ | | |
| | $\mathrm{CI}_l$ | $\mathrm{CI}_u$ | $\hat{\kappa}$ | $\mathrm{CI}_l$ | $\mathrm{CI}_u$ | $\hat{\kappa}$ | $\mathrm{CI}_l$ | $\mathrm{CI}_u$ | $\hat{\pi}$ |
| $V(d_0)$ | 0.453 | 0.672 | 0.567 | 0.471 | 0.699 | 0.591 | 0.466 | 0.700 | 0.589 |
| $V(d_1)$ | 0.699 | 0.857 | 0.784 | 0.711 | 0.870 | 0.798 | 0.710 | 0.870 | 0.797 |
| $V(d_2^2)$ | 0.834 | 0.948 | 0.898 | 0.834 | 0.948 | 0.898 | 0.834 | 0.948 | 0.898 |
| Hubert† | 0.453 | 0.672 | 0.567 | 0.273 | 0.564 | 0.424 | 0.271 | 0.564 | 0.423 |

† Hubert disagreement equals the nominal disagreement $V(d_0)$ when $g = 2$.

Table 4 summarizes the relevant aspects of the data. According to the Hubert distance the raters disagree a lot, as only $5$ items have a distance of $0$ and the rest a distance of one. On th other hand, $V(d_0)$ equals $6$ minus the maximal agreement (i.e., the number of elements distinct from the mode), and results in much smaller chance-adjusted agreement.

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**Example 8.** Zapf et al. (2016) studied bootstrap intervals for Fleiss' kappa and Krippendorff's alpha using simulations and a case study. Their case study concerned histopathological assessment of breast cancer, and involved ratings done by $R = 4$ senior pathologists and $n = 50$ breast cancer biopsies. We apply the arcsine method to calculate confidence intervals and point estimates (Table 5). We focus on Cohen's kappa since the same four pathologists rate each cancer biopsy, but we include a column for Fleiss' kappa when $g = 4$ for comparisons sake. When $g = 4$, Cohen's kappa and Fleiss' kappa are as good as indistinguishable. As can be verified using the code in the supplementary material, this happens for the other $g$s as well. It is not generally the case that Fleiss' kappa and Cohen's kappa nearly coincide, but is likely to happen if the marginal ratings are approximately the same for all raters, as is the case in this data set. There is a sizable difference between the disagreement functions, but there is not typically a big difference when changing $g$s, provided we keep the disagreement functions constant. It remains to be seen if this common or not. The exception is Hubert's disagreement function, which decreases by quite a bit. (As in the Fleiss (1971) example, this is expected, as the Hubert's disagreement function is a consensus measure.) Observe that the quadratic Fréchet variance $V(d_2^2)$ does not change with $g$ at all. The estimators are not exactly the same for the Cohen type chance agreement, but one can show that the disagreement at $x$ equals with $g = 2 + i$ equals $i$ times the disagreement when $g = 2$. This implies that Fleiss' kappa is invariant under $g$.

# 5 Simulations

## 5.1 Simulation of confidence sets when $g = 2$

We include a small simulation study about the performance of confidence sets using two models.

(i) **Number of raters** $R$. We use $2, 5, 20$, corresponding to a small, medium, and large selection of raters.

(ii) **Sample sizes** $n$. We use $n = 10, 40, 100$, corresponding to small, medium-sized, and large agreement studies.

(iii) **Disagreement functions**. We use the classics: The nominal disagreement $1[x \neq y]$, quadratic disagreement $(x - y)^2$, and absolute value disagreement $|x - y|$.

(iv) **Models.** We simulate from two models. A Perreault–Leigh model for discrete ratings and a normal model for continuous ratings. These models are described in more detail below.

(v) **Methods.** We will use three methods for confidence interval construction: A basic interval without transformations, an arcsine-transformed interval, and a Fisher-transformed interval.

We describe our three confidence interval constructions only for Cohen's kappa, as the intervals using Fleiss' kappa can be found by replacing every instance $\hat{\kappa}_d$ with $\hat{\pi}_d$ and $\hat{\sigma}_\kappa^2$ with $\hat{\sigma}_\pi^2$. We use two-sided $t$-distribution-based confidence intervals with nominal level $1 - \alpha = 0.95$. The basic interval is

$$[\hat{\kappa}_d - t_{1-\alpha/2}(n-1)\hat{\sigma}_\kappa, \hat{\kappa}_d + t_{1-\alpha/2}(n-1)\hat{\sigma}_\kappa], \tag{5.1}$$

where $\hat{\sigma}(\hat{\kappa}_d)$ is the estimated variance described in equation (4.9) and $t_\alpha(\nu)$ is the $\alpha$-percentile of the $t$-distribution with $\nu$ degrees of freedom.

In the arcsine interval replaces the basic limits with

$$\sin\left(\arcsin \hat{\kappa}_d \pm t_{1-\alpha/2}(n-1)(1-\hat{\kappa}_d^2)^{-1}\hat{\sigma}_\kappa^2\right), \tag{5.2}$$

where $(1-\hat{\kappa}_d^2)^{-1}\hat{\sigma}_\kappa^2$ is the asymptotic variance of $\arcsin \hat{\kappa}_d$ (4.11). The Fisher interval uses the area hyperbolic tangent,

$$\tanh\left(\operatorname{artanh} \hat{\kappa}_d \pm t_{1-\alpha/2}(n-1)(1-\hat{\kappa}_d^2)^{-2}\hat{\sigma}_\kappa^2\right), \tag{5.3}$$

where $(1-\hat{\kappa}_d^2)^{-2}\hat{\sigma}_\kappa^2$ is the asymptotic variance of $\operatorname{artanh} \hat{\kappa}_d$ (4.12).

### 5.1.1 A Perreault–Leigh model

Perreault and Leigh (1989) discusses a particular model for ratings where each rater either knows the correct answer or guesses uniformly at random. Similar models have been used by Gwet (2008); Maxwell (1977), among others. We assume there are five categories encoded as $C = (-2, -1, 0, 1, 2)$, and the distribution of the true classification distribution is uniform. For each item rated, the $j$th rater knows the correct classification with probability $\sqrt{0.8}$. If not, he guesses, picking a number from $C$ uniformly at random. Then $\kappa_d = \pi_d = 0.8$ for all weights and number of raters, as can be verified by following the arguments of Perreault and Leigh (1989). We run each simulation $N = 10,000$ times.

The simulated lengths and coverages for Cohen's kappa are in Table 6. The table for Fleiss' kappa is similar, and can be found in the appendix, p. 24. Two features stand out in Table 6. First, the confidence intervals have almost indistinguishable lengths and coverages when either $R$ or $n$ is large. Second, the basic interval has worse coverage than the arcsine and Fisher intervals when $n$ is small, with the Fisher interval having slightly better coverage than the arcsine interval. However, the better coverage comes at the expense of greater lengths. In particular, for the absolute value weight, the coverage of the arcsine interval is greater than the coverage of the Fisher interval, but its length is shorter.

Table 6: Coverage (first entry) and lengths (second entry) of confidence intervals: Perreault–Leigh model, Cohen's kappa

| | | Perreault–Leigh model, Cohen's kappa | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | **Method** | $R=2$ | | | $R=5$ | | | $R=20$ | | |
| | $n$ | 10 | 40 | 100 | 10 | 40 | 100 | 10 | 40 | 100 |
| **Weights** | | | | | | | | | | |
| Nominal | Basic | 0.81 | **0.96** | **0.96** | 0.92 | **0.95** | **0.95** | **0.96** | **0.95** | **0.95** |
| | | 0.53 | 0.30 | 0.18 | 0.41 | 0.18 | 0.11 | 0.23 | 0.09 | 0.06 |
| | Arcsine | **0.98** | **0.95** | **0.95** | **0.97** | **0.95** | **0.95** | **0.95** | **0.95** | **0.95** |
| | | 0.73 | 0.29 | 0.18 | 0.43 | 0.18 | 0.11 | 0.23 | 0.09 | 0.06 |
| | Fisher | **0.97** | **0.96** | **0.96** | **0.96** | **0.95** | **0.95** | **0.95** | 0.94 | **0.95** |
| | | 0.91 | 0.32 | 0.19 | 0.50 | 0.19 | 0.11 | 0.24 | 0.09 | 0.06 |
| Quadratic | Basic | 0.65 | 0.87 | 0.92 | 0.84 | 0.93 | **0.95** | 0.94 | **0.95** | **0.95** |
| | | 0.58 | 0.39 | 0.26 | 0.49 | 0.26 | 0.16 | 0.34 | 0.14 | 0.08 |
| | Arcsine | 0.82 | 0.89 | 0.93 | 0.88 | 0.94 | **0.95** | 0.94 | **0.95** | **0.95** |
| | | 0.78 | 0.39 | 0.25 | 0.55 | 0.26 | 0.16 | 0.34 | 0.14 | 0.08 |
| | Fisher | **0.95** | 0.91 | **0.95** | 0.90 | 0.94 | **0.95** | 0.93 | **0.95** | **0.95** |
| | | 0.94 | 0.44 | 0.27 | 0.65 | 0.27 | 0.16 | 0.37 | 0.14 | 0.08 |
| Absolute value | Basic | 0.80 | 0.91 | 0.93 | 0.90 | 0.94 | **0.95** | **0.95** | **0.95** | **0.95** |
| | | 0.55 | 0.33 | 0.21 | 0.44 | 0.21 | 0.13 | 0.27 | 0.11 | 0.06 |
| | Arcsine | **0.98** | 0.94 | **0.95** | 0.94 | **0.95** | **0.95** | **0.95** | **0.95** | **0.95** |
| | | 0.75 | 0.33 | 0.21 | 0.47 | 0.21 | 0.13 | 0.27 | 0.11 | 0.06 |
| | Fisher | **0.97** | **0.95** | **0.95** | **0.95** | **0.95** | **0.96** | 0.94 | **0.95** | **0.95** |
| | | 0.93 | 0.35 | 0.21 | 0.55 | 0.21 | 0.13 | 0.28 | 0.11 | 0.07 |

Coverages greater than $0.95$ are in bold.

### 5.1.2 Normal model

In this study, the rating data is distributed according to the multivariate normal $N(0,\Sigma)$, where $\Sigma$ is the $R\times R$ correlation matrix with off-diagonal elements $\Sigma_{j_1j_2}=\rho$. Recall that $d_1$ denotes the absolute value disagreement and $d_2^2$ the quadratic disagreement. The true values are $\kappa_{d_2}=\pi_{d_2^2}=\rho$ and $\kappa_{d_1}=\pi_{d_1}=1-\sqrt{1-\rho}$. See the appendix (p. 21) for details on the computation of these true value. We use $\rho=0.7$, hence $\kappa_{d_2^2}=0.7$ and $\kappa_{d_1}=0.45$. Note that the nominal weight will not work for this continuous data, hence we simulate using only the quadratic and absolute value weights. We run each simulation $N=1,000$ times[4]. We note that agreement coefficients are frequently called concordance coefficients when dealing with continuous data. Lin's concordance coefficient (Lin, 1989) is a prominent example.

The simulated lengths and coverages for Cohen's kappa are in Table 7. The table for Fleiss' kappa is similar, and can be found in the appendix (p. 24). From the table below, we see that there is barely any difference between the three confidence interval constructions. Taken together with the results for the Perreault–Leigh model, where the basic interval performs worse than the other two, we would recommend the usage of either the arcsine or Fisher interval.

[4]We use fewer simulations $(1,000)$ than in the previous simulation $(10,000)$ since estimation is far more computationally expensive when dealing with continuous data, as it does not allow for binning.

Table 7: Coverage (first entry) and lengths (second entry) of confidence intervals: Normal model, Cohen's kappa

| | | | **Cohen's kappa: Normal model** | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | **Method** | | $R = 2$ | | | $R = 5$ | | | $R = 20$ | | |
| | | $n$ | 10 | 40 | 100 | 10 | 40 | 100 | 10 | 40 | 100 |
| **Weights** | | | | | | | | | | | |
| Quadratic | Basic | | 0.88 | 0.92 | **0.95** | 0.91 | **0.95** | 0.94 | 0.88 | 0.93 | **0.95** |
| | | | 0.66 | 0.32 | 0.20 | 0.50 | 0.23 | 0.14 | 0.43 | 0.20 | 0.12 |
| | Arcsine | | 0.88 | 0.93 | **0.95** | 0.90 | **0.95** | 0.94 | 0.87 | 0.92 | 0.94 |
| | | | 0.67 | 0.32 | 0.20 | 0.49 | 0.23 | 0.14 | 0.42 | 0.20 | 0.12 |
| | Fisher | | 0.90 | 0.94 | 0.94 | 0.88 | 0.94 | 0.94 | 0.86 | 0.92 | 0.94 |
| | | | 0.70 | 0.33 | 0.20 | 0.51 | 0.23 | 0.14 | 0.43 | 0.20 | 0.12 |
| Absolute value | Basic | | 0.92 | 0.94 | 0.94 | 0.92 | 0.93 | **0.95** | 0.87 | 0.94 | 0.94 |
| | | | 0.67 | 0.31 | 0.19 | 0.46 | 0.21 | 0.13 | 0.38 | 0.18 | 0.11 |
| | Arcsine | | 0.93 | 0.94 | 0.94 | 0.92 | 0.93 | **0.95** | 0.87 | 0.94 | 0.94 |
| | | | 0.65 | 0.31 | 0.19 | 0.45 | 0.21 | 0.13 | 0.38 | 0.18 | 0.11 |
| | Fisher | | 0.93 | 0.94 | **0.95** | 0.92 | 0.93 | **0.95** | 0.86 | 0.94 | 0.94 |
| | | | 0.65 | 0.31 | 0.19 | 0.45 | 0.21 | 0.13 | 0.38 | 0.18 | 0.11 |

Coverages greater than $0.95$ are in bold.

### 5.2 Simulation of confidence sets when $g \neq 2$

Table 8 contains simulations from the Perreault–Leigh model (p. 14) with $N = 1000$ repetitions and $R = 5$ raters using the Fréchet variances $V(d_0)$, $V(d_1)$, $V(d_2^2)$, and Hubert's disagreement function. To save space, we drop the basic confidence interval in the simulation. As before, we show the results only for the Cohen-type disagreement, with the Fleiss-type disagreement relegated to the appendix (p. 25). The methods have virtually indistinguishable properties, with very similar coverages and lengths across the board. All coverages are decent, with the exception of the quadratic Fréchet variance with small $n$, where the coverage drops into the high 80s.

## 6 Concluding remarks

When choosing an agreement coefficient one has to carefully think through exactly what one wishes to measure. The Fréchet variances are attractive due to their interpretation. You measure how much do the raters disagree with the (generalized) mean rater, and then adjust for chance. In the case of nominal data, we measure disagreement with the modal rater. When dealing with numerical data, we may measure disagreement with the median rater (using the absolute value distance), or the mean rater (using the quadratic distance), or use any other Fréchet variance defined on numeric data.

When dealing with nominal data, we believe that using the Fréchet variance that measures the distance from the mode is a reasonable choice. But other options are certainly possible, even when dealing with $g$-wise agreement measures. For instance, one could use the entropy instead, with distance measure $d(x_1, x_2, \ldots, x_g) = -\sum_{i=1}^{g} \frac{\#i}{g} \log \frac{\#i}{g}$,where $\#i$ counts the number of elements in $(x_1, x_2, \ldots, x_g)$ clas-

Table 8: Coverage (first entry) and lengths (second entry) of confidence intervals: Perreault–Leigh model, Cohen's kappa

| | | **Perreault–Leigh model, Cohen's kappa** | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | **Method** | $g = 3$ | | | $g = 4$ | | | $g = 5$ | | |
| | $n$ | 10 | 40 | 100 | 10 | 40 | 100 | 10 | 40 | 100 |
| **Weights** | | | | | | | | | | |
| $V(d_0)$ | Arcsine | **0.98** | **0.96** | 0.94 | **0.97** | **0.95** | 0.94 | **0.98** | 0.93 | 0.93 |
| | | 0.41 | 0.16 | 0.10 | 0.39 | 0.16 | 0.10 | 0.38 | 0.15 | 0.09 |
| | Fisher | **0.96** | **0.96** | 0.94 | **0.96** | **0.96** | 0.94 | **0.96** | 0.94 | 0.94 |
| | | 0.46 | 0.17 | 0.10 | 0.44 | 0.16 | 0.10 | 0.42 | 0.15 | 0.09 |
| $V(d_1)$ | Arcsine | 0.94 | 0.94 | **0.95** | 0.94 | 0.94 | 0.94 | 0.94 | **0.95** | **0.95** |
| | | 0.49 | 0.21 | 0.13 | 0.45 | 0.19 | 0.11 | 0.44 | 0.19 | 0.11 |
| | Fisher | **0.96** | 0.94 | **0.95** | **0.95** | **0.95** | **0.95** | **0.96** | **0.95** | **0.95** |
| | | 0.55 | 0.21 | 0.13 | 0.51 | 0.19 | 0.12 | 0.51 | 0.19 | 0.12 |
| $V(d_2^2)$ | Arcsine | 0.90 | 0.94 | **0.96** | 0.88 | 0.94 | 0.94 | 0.88 | 0.93 | 0.94 |
| | | 0.56 | 0.25 | 0.16 | 0.56 | 0.26 | 0.16 | 0.57 | 0.25 | 0.16 |
| | Fisher | 0.92 | 0.94 | **0.96** | 0.91 | **0.95** | **0.95** | 0.89 | 0.94 | **0.95** |
| | | 0.65 | 0.27 | 0.16 | 0.65 | 0.27 | 0.16 | 0.66 | 0.27 | 0.16 |
| Hubert | Arcsine | **0.98** | **0.95** | **0.96** | **0.98** | **0.95** | **0.96** | **0.98** | **0.95** | **0.95** |
| | | 0.52 | 0.22 | 0.13 | 0.62 | 0.26 | 0.16 | 0.71 | 0.31 | 0.19 |
| | Fisher | **0.97** | **0.96** | **0.96** | **0.97** | **0.96** | **0.96** | **0.98** | **0.96** | 0.94 |
| | | 0.57 | 0.22 | 0.13 | 0.67 | 0.27 | 0.16 | 0.77 | 0.31 | 0.19 |

Coverages greater than $0.95$ are in bold.

sified as $i$. The topic of how to choose reasonable distance measures for $g$-wise agreement studies has not been thoroughly studied, and there might be options preferable to the Fréchet variances that have not yet been found.

We have only covered rectangular design, where every item is rated by the same number of raters. It is quite easy to generalize the definitions of $\kappa_d$ and $\pi_d$ to non-rectangular designs, as we have done in the appendix, p. 22. But inference appears to be quite difficult, probably requiring additional assumptions for the case of non-exchangeable ratings.

The confidence intervals based on the arcsine and Fisher transforms perform better than the basic, untransformed interval. It is unclear which one of these intervals to prefer, but it barely matters when the sample size is sufficiently large. It might be possible to improve all of these intervals. Small-sample corrections to the variance appears feasible, with potential openings in the in the application of the delta rule and in the calculation of $\Sigma$ of Lemma 9. We have used the arcsine and Fisher transforms to improve approximate normality of $\hat{\kappa}_d$ and $\hat{\pi}_d$, but this choice is semi-arbitrary. Better variance-stabilizing transformations might be found by inspecting the formula for the variances of $\hat{\kappa}_d$ and $\hat{\pi}_d$ in Proposition 5. The confidence intervals used in the simulation are only known to be first-order accurate. To make second-order accurate confidence intervals, it would be possible to use the explicit formula for the variances to construct studentized confidence intervals, i.e., bootstrap-$t$ intervals (Efron, 1987), which are second-order accurate.

None of these approaches are guaranteed to help when $n$ is small, especially when dealing with categorical data, as the sampling distributions of $\hat{\kappa}_d$ and $\hat{\pi}_d$ are discrete and highly irregular. For example, consider

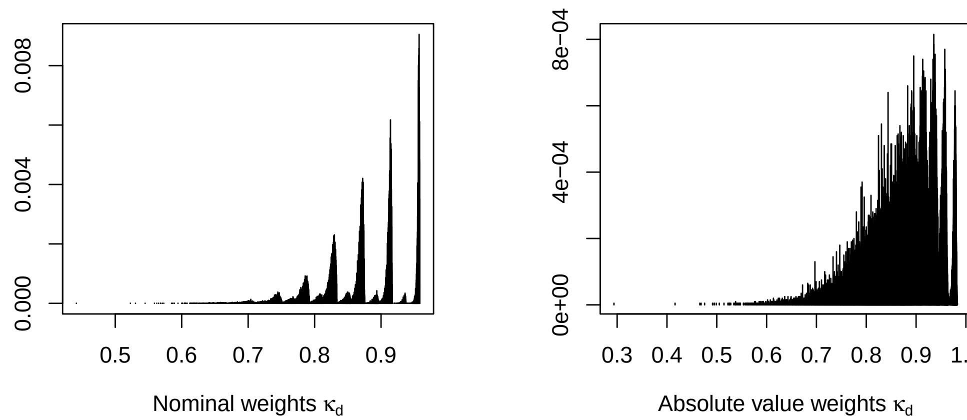


Figure 6.1: Sample distribution of $\hat{\kappa}_d$ for nominal (left) and absolute value (right) weights. Both plots omit a dominating spike at 1. Here $n = 20$ and $j = 5$, and we use the Perreault–Leigh model (same parameters as in section 5.1) to simulate the data. There were 2573 unique values for the nominal weight and 8790 unique values for the absolute value weight after $N = 200,000$ simulations.

the sample distribution of the Perreault–Leigh model (section 5.1) when $n = 20$ and $R = 20$, displayed in Figure 6.1. (We omit a dominating spike at 1.) As there are $C = 5 < \infty$ categories, there is a finite number of possible values for $\hat{\kappa}_d$ to take, which is strongly reflected in the plots, especially for the nominal weight.

The superior performance of methods such as the bootstrap-$t$ depend on the quantity $\frac{\hat{\theta}-\theta}{\text{se}}$ being approximately pivotal, i.e., approximately the same for all parameters, possibly after applying a transformation. Judging from the plots above, there is no such transformation.

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# Appendix

### Agreement vs disagreement

Agreement weighting functions are frequently standardized to guarantee that $w(x_1, x_2) \geq 0$, e.g., $w(x_1, x_2) = 1 - |x_1 - x_2| / \max(|x_1 - x_2|)$ for the absolute value weights. The standardization is not necessary, as they do not change the value of $\kappa_d$ and $\pi_d$ when it is possible (i.e., when $\max(|x_1 - x_2|) < \infty$), and is undefined otherwise. We choose not use this operation, as it does not change the value of the agreement coefficients in this paper, and is impossible to do when the range of $x_1, x_2$ is unbounded.

**Proof of equivalence between $V(d_p)(\boldsymbol{x}_1, \boldsymbol{x}_2)$ and $||\boldsymbol{x}_1 - \boldsymbol{x}_2||$**

*Proof.* We will show that

$$V(d_p)[\boldsymbol{x}_1, \boldsymbol{x}_2] = \frac{1}{2}||\boldsymbol{x}_1 - \boldsymbol{x}_2||_p, \quad V(d_p^p)[\boldsymbol{x}_1, \boldsymbol{x}_2] = \frac{1}{2^p}||\boldsymbol{x}_1 - \boldsymbol{x}_2||_p^p.$$

First, consider the case when $p \geq 1$. Using translation invariance and homogeneity of the norm,

$$\begin{aligned}
& ||x_1 - \mu||_p + ||x_2 - \mu||_p, \\
= \quad & ||x_1 - \frac{x_1 + x_2}{2} - \mu + \frac{x_1 + x_2}{2}||_p + ||x_2 - \frac{x_1 + x_2}{2} - \mu + \frac{x_1 + x_2}{2}||_p, \\
= \quad & ||\frac{x_1 - x_2}{2} - \nu||_p + || - \frac{x_1 - x_2}{2} - \nu||_p, \\
= \quad & ||a - \nu||_p + ||a + \nu||_p,
\end{aligned}$$

where $a = \frac{x_1 - x_2}{2}$ and $\nu = \mu - \frac{x_1 + x_2}{2}$.

Observe that

$$\text{argmin}_\nu \, ||a + \nu||_p + ||a - \nu||_p = 0, \quad \text{for all } a$$

implies $\mu = \frac{x_1 + x_2}{2}$.

By the Minkowski inequality,

$$2^p||a||^p = ||a + \nu + a - \nu||^p \leq (||a - \nu|| + ||a + \nu||)^p.$$

This is an equality if $||a - \nu|| = ||a + \nu|| = ||a||$, i.e., when $\nu = 0$, as the left side equals $(||a - \mu|| + ||a + \mu||)^p = 2^p||a||^p$. Now it's easy to verify that $V(d_p)$ and $V(d_p^p)$ have the claimed form; just substitute the value $\mu = \frac{x_1 + x_2}{2}$ into the formula for the Fréchet variance, $\frac{1}{2}(||x_1 - \mu||_p + ||x_2 - \mu||_p)$.

When $0 < p < 1$, the function $\mu \mapsto ||x_1 - \mu||_p + ||x_2 - \mu||_p$ is stepwise concave on $[-\infty, x_1]$,$[x_1, x_2]$, and $[x_2, \infty)$, hence its minimum is either $x_1, x_2$, or both. It's clear that both $x_1$ and $x_2$ maps to $||x_1 - x_2||_p$, hence both are Fréchet means. The case $p = 0$ is obvious and omitted. □

**True values in the normal simulation**

We give a brief explanation why the true values of $\kappa_d$ and $\pi_d$ are 0.8 for the quadratic weights and $1 - \sqrt{0.2}$ for the absolute value weights.

First notice that, since the marginals of $X_{j_1}$ and $X_{j_2}$ are equal for all $j_1, j_2$, we have that $\kappa_d = \pi_d$. Moreover, we can ignore the number of raters, since the pairwise distribution do not depend on them. Then, from standard theory about the multivariate and folded normal, we find that

$$E(|X_{j_1} - X_{j_2}|) = 2\sqrt{\frac{1 - \rho}{\pi}}, \quad E(|X_{j_1} - X_{j_2}|^2) = 2(1 - \rho).$$

Let $X'_{j_1}$ be a copy of $X_{j_1}$ that is independent of $X_{j_2}$. Then $E(|X'_{j_1} - X_{j_2}|) = 2/\sqrt{\pi}$ and $E(|X'_{j_1} - X_{j_2}|^2) = 2$. Now rewrite the kappas using disagreement instead of agreement. Use the fact that $(p_{wa} - p_{fa})/(1 - p_{fa}) = 1 - d_{wa}/d_{fa}$, where $d_{wa} = 1 - E(w(X_{j_1}, X_{j_2}))$ and $d_{fa} = 1 - E(w(X'_{j_1}, X_{j_2}))$, where $X'_{j_1}$ is a copy of $X_{j_1}$ that is independent of $X_{j_2}$.

Thus $\kappa_d = \pi_d = 1 - E(|X_{j_1} - X_{j_2}|)/E(|X'_{j_1} - X_{j_2}|^2) = 1 - \sqrt{1 - \rho}$ for the absolute value weights and $1 - E(|X_{j_1} - X_{j_2}|^2)/E(|X'_{j_1} - X_{j_2}|^2) = \rho$ for the quadratic weights.

### Variance of $U$-statistics

Let $U_n^1$ and $U_n^2$ be two $U$-statistics of $n$ observations with symmetric kernel functions $\psi_1$, $\psi_2$ of dimension $k_1$ and $k_2$. Define

$$\sigma_{cc}^2 = \operatorname{Cov}(E[\psi_1(X_1,\dots,X_{k_1}) \mid X_1,\dots,X_c)], E[\psi_2(X_1,\dots,X_{k_2}) \mid X_1,\dots,X_c)]).$$

**Proposition 9.** *The exact covariance of $U_1^n$ and $U_2^n$ is*

$$\operatorname{Cov}(U_1^n, U_2^n) = \binom{n}{k_1}^{-1} \sum_{c=1}^{k_1} \binom{k_2}{c}\binom{n-k_2}{k_1-c}\sigma_{cc}^2.$$

*If $k_1$ and $k_2$ are fixed, its asymptotic variance is $n \operatorname{Cov}(U_1^n, U_2^n) \to k_1 k_2 \sigma_{11}^2$.*

*Proof.* See (Lee, 2019, Theorem 2, p. 17) and (Lee, 2019, Theorem 2, p. 76) for a proof. □

### Expanding the definitions

Here is sketch of how we could expand the definitions in Section 2 to encompass more complicated scenarios. We restrict ourselves to $g = 2$, but the analysis can be expanded to arbitrary $g$. Suppose that any infinite number of raters $R$ is possible, the raters are not exchangeable, and that not every item is rated by every rater.

Let $X$ denote a rating, $J$ be a raters, and $I$ be the items rated. Suppose we sample pairs $(X_1, J_1, I_1), (X_2, J_2, I_2)$ independently from the same distribution $F$. Then we may define

$$\begin{aligned} D_d &= E[d(X_1, X_2) \mid I_1 = I_2, J_1 \neq J_2], \\ C_d &= E[d(X_1, X_2) \mid J_1 \neq J_2], \\ F_d &= E[d(X_1, X_2)]. \end{aligned} \tag{6.1}$$

These quantities have natural sample analogues, e.g.,

$$\hat{D}_d = N^{-1} \sum_{i=1}^{n} \sum_{j_1 \neq j_2} d(x_{ij_1}, x_{ij_2}),$$

where $N$ is the total number of paired observations and the rater indices run over the raters who observed at the $i$th observation $x$. Population and sample definitions of Cohen's kappa and Fleiss' kappa follow as laid out in the main text, e.g. $\kappa_d = 1 - D_d/C_d$.

### Krippendorff's alpha

Now suppose that the ratings can take on only a finite number $C$ distinct values. Define $o_{ck}$ as the number of times a pair of raters has classified an item into $c$ and $k$, i.e.,

$$o_{ck} = \sum_{i=1}^{n} \sum_{j_1 \neq j_2} 1[x_{ij_1} = c, x_{ij_2} = k].$$

Then $N = \sum_{c,k} o_{ck}$ and $\hat{D}_d = N^{-1} \sum_{c,k} o_{ck} d(c,k)$. Moreover, define $n_c$ as the number of items classified as $c$. Then $n_c = \sum_k o_{ck}$, $\sum_c n_c = N$, and $\sum_{c,k} n_c n_k d(c,k) = N^2 \hat{F}_d$.

**Proposition 10.** *Using the definitions above, $\hat{\alpha}_d = \hat{\pi}_d + \frac{1}{N}(1 - \hat{\pi}_d)$. Since there are $N = 2Rn$ rating pairs in the rectangular setup used in Section 2, $\hat{\alpha}_w = \hat{\pi}_w + \frac{1}{2Rn}(1 - \hat{\pi}_w)$ in that case.*

*Proof.* The definition of $\hat{\alpha}_w$ can be found on Krippendorff (2018, p .235),

$$\hat{\alpha}_w = 1 - (N-1)\frac{\sum_{c\neq k} o_{ck} d(c,k)}{\sum_{c\neq k} n_c n_k d(c,k)}.$$

From the definitions above, and the fact that $d(c,k) = 0$ when $c = k$, we find that

$$\sum_{c\neq k} o_{ck} d(c,k) = \sum_{c,k} o_{ck} d(c,k) = N\hat{D}_d.$$

In the same way,

$$\sum_{c\neq k} n_c n_k d(c,k) = \sum_{c,k} n_c n_k d(c,k) = N^2\hat{F}_d.$$

Thus

$$\hat{\alpha}_d = 1 - \frac{(N-1)}{N}\frac{\hat{D}_d}{\hat{F}_d} = 1 - \frac{\hat{D}_d}{\hat{F}_d} + \frac{1}{N}\frac{\hat{D}_d}{\hat{F}_d},$$

and using that $\hat{\pi}_d = 1 - \frac{\hat{D}_d}{\hat{F}_d}$, we're done. □

### Simulation of Fleiss' kappa

Here are the results of the simulation study in 5.1 for Fleiss' kappa.

Table 9: Coverage (first entry) and lengths (second entry) of confidence intervals: Perreault–Leigh model, Fleiss' kappa

| | | | **Fleiss' kappa: Perreault–Leigh model** | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | **Method** | | | $R=2$ | | | $R=5$ | | | $R=20$ | |
| | | $n$ | 10 | 40 | 100 | 10 | 40 | 100 | 10 | 40 | 100 |
| **Weights** | | | | | | | | | | | |
| Nominal | Basic | | 0.82 | 0.95 | 0.95 | 0.93 | 0.95 | 0.95 | 0.96 | 0.95 | 0.96 |
| | | | 0.55 | 0.30 | 0.18 | 0.41 | 0.18 | 0.11 | 0.23 | 0.09 | 0.06 |
| | Arcsine | | 0.98 | 0.94 | 0.94 | 0.98 | 0.95 | 0.95 | 0.95 | 0.95 | 0.95 |
| | | | 0.76 | 0.30 | 0.18 | 0.44 | 0.18 | 0.11 | 0.23 | 0.09 | 0.06 |
| | Fisher | | 0.97 | 0.96 | 0.96 | 0.96 | 0.96 | 0.95 | 0.95 | 0.94 | 0.95 |
| | | | 0.95 | 0.32 | 0.19 | 0.51 | 0.19 | 0.11 | 0.24 | 0.09 | 0.06 |
| Quadratic | Basic | | 0.65 | 0.86 | 0.92 | 0.85 | 0.93 | 0.94 | 0.94 | 0.95 | 0.95 |
| | | | 0.60 | 0.39 | 0.26 | 0.50 | 0.26 | 0.16 | 0.34 | 0.14 | 0.08 |
| | Arcsine | | 0.83 | 0.89 | 0.93 | 0.89 | 0.94 | 0.94 | 0.94 | 0.95 | 0.95 |
| | | | 0.82 | 0.39 | 0.25 | 0.56 | 0.26 | 0.16 | 0.34 | 0.14 | 0.08 |
| | Fisher | | 0.96 | 0.91 | 0.94 | 0.91 | 0.94 | 0.95 | 0.93 | 0.95 | 0.95 |
| | | | 0.98 | 0.44 | 0.27 | 0.67 | 0.27 | 0.16 | 0.37 | 0.14 | 0.08 |
| Absolute value | Basic | | 0.81 | 0.92 | 0.94 | 0.91 | 0.95 | 0.95 | 0.95 | 0.95 | 0.95 |
| | | | 0.57 | 0.33 | 0.21 | 0.44 | 0.21 | 0.13 | 0.27 | 0.11 | 0.06 |
| | Arcsine | | 0.99 | 0.93 | 0.95 | 0.94 | 0.95 | 0.95 | 0.95 | 0.95 | 0.95 |
| | | | 0.79 | 0.33 | 0.21 | 0.48 | 0.21 | 0.13 | 0.27 | 0.11 | 0.06 |
| | Fisher | | 0.97 | 0.95 | 0.95 | 0.96 | 0.95 | 0.95 | 0.94 | 0.95 | 0.95 |
| | | | 0.97 | 0.36 | 0.21 | 0.56 | 0.21 | 0.13 | 0.28 | 0.11 | 0.07 |

Table 10: Coverage (first entry) and lengths (second entry) of confidence intervals: Normal model, Fleiss' kappa

| | | | **Fleiss' kappa: Normal model** | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | **Method** | | | $R=2$ | | | $R=5$ | | | $R=20$ | |
| | | $n$ | 10 | 40 | 100 | 10 | 40 | 100 | 10 | 40 | 100 |
| **Weights** | | | | | | | | | | | |
| Quadratic | Basic | | 0.89 | 0.93 | 0.95 | 0.90 | 0.95 | 0.94 | 0.90 | 0.93 | 0.95 |
| | | | 0.70 | 0.33 | 0.20 | 0.50 | 0.23 | 0.14 | 0.42 | 0.20 | 0.12 |
| | Arcsine | | 0.90 | 0.93 | 0.95 | 0.90 | 0.95 | 0.94 | 0.90 | 0.92 | 0.94 |
| | | | 0.71 | 0.32 | 0.20 | 0.49 | 0.23 | 0.14 | 0.42 | 0.20 | 0.12 |
| | Fisher | | 0.92 | 0.93 | 0.95 | 0.89 | 0.94 | 0.94 | 0.88 | 0.92 | 0.94 |
| | | | 0.74 | 0.33 | 0.20 | 0.51 | 0.23 | 0.14 | 0.43 | 0.20 | 0.12 |
| Absolute value | Basic | | 0.92 | 0.95 | 0.95 | 0.91 | 0.96 | 0.93 | 0.90 | 0.93 | 0.93 |
| | | | 0.71 | 0.32 | 0.20 | 0.47 | 0.21 | 0.13 | 0.39 | 0.18 | 0.11 |
| | Arcsine | | 0.92 | 0.95 | 0.95 | 0.90 | 0.95 | 0.92 | 0.89 | 0.93 | 0.93 |
| | | | 0.69 | 0.32 | 0.20 | 0.46 | 0.21 | 0.13 | 0.39 | 0.18 | 0.11 |
| | Fisher | | 0.92 | 0.95 | 0.95 | 0.90 | 0.95 | 0.93 | 0.89 | 0.93 | 0.93 |
| | | | 0.68 | 0.32 | 0.20 | 0.46 | 0.21 | 0.13 | 0.39 | 0.18 | 0.11 |

Table 11: Coverage (first entry) and lengths (second entry) of confidence intervals: Perreault–Leigh model, Fleiss' kappa ($R = 5$)

| | | | **Perreault–Leigh model, Fleiss' kappa** | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | **Method** | | $g = 3$ | | | $g = 4$ | | | $g = 5$ | | |
| | | $n$ | 10 | 40 | 100 | 10 | 40 | 100 | 10 | 40 | 100 |
| **Weights** | | | | | | | | | | | |
| $V(d_0)$ | Arcsine | | 0.97 | 0.94 | 0.96 | 0.98 | 0.95 | 0.95 | 0.98 | 0.95 | 0.94 |
| | | | 0.41 | 0.16 | 0.10 | 0.39 | 0.16 | 0.10 | 0.38 | 0.15 | 0.09 |
| | Fisher | | 0.96 | 0.94 | 0.96 | 0.96 | 0.94 | 0.95 | 0.96 | 0.95 | 0.94 |
| | | | 0.46 | 0.17 | 0.10 | 0.44 | 0.16 | 0.10 | 0.42 | 0.15 | 0.09 |
| $V(d_1)$ | Arcsine | | 0.93 | 0.95 | 0.95 | 0.94 | 0.94 | 0.96 | 0.94 | 0.95 | 0.96 |
| | | | 0.51 | 0.21 | 0.13 | 0.45 | 0.19 | 0.12 | 0.45 | 0.19 | 0.11 |
| | Fisher | | 0.95 | 0.94 | 0.95 | 0.96 | 0.94 | 0.96 | 0.96 | 0.95 | 0.96 |
| | | | 0.57 | 0.21 | 0.13 | 0.51 | 0.19 | 0.12 | 0.51 | 0.19 | 0.12 |
| $V(d_2^2)$ | Arcsine | | 0.88 | 0.94 | 0.94 | 0.89 | 0.94 | 0.95 | 0.90 | 0.94 | 0.95 |
| | | | 0.56 | 0.26 | 0.16 | 0.59 | 0.26 | 0.16 | 0.58 | 0.26 | 0.16 |
| | Fisher | | 0.90 | 0.95 | 0.94 | 0.90 | 0.94 | 0.95 | 0.92 | 0.94 | 0.95 |
| | | | 0.67 | 0.27 | 0.16 | 0.68 | 0.27 | 0.16 | 0.67 | 0.27 | 0.16 |
| Hubert | Arcsine | | 0.97 | 0.95 | 0.95 | 0.98 | 0.96 | 0.95 | 0.97 | 0.95 | 0.96 |
| | | | 0.53 | 0.22 | 0.13 | 0.62 | 0.26 | 0.16 | 0.70 | 0.31 | 0.19 |
| | Fisher | | 0.96 | 0.96 | 0.95 | 0.98 | 0.96 | 0.95 | 0.97 | 0.96 | 0.96 |
| | | | 0.57 | 0.22 | 0.13 | 0.67 | 0.27 | 0.16 | 0.77 | 0.31 | 0.19 |